

MPC for Robot Manipulators With Integral Sliding Modes Generation

Gian Paolo Incremona, *Member, IEEE*, Antonella Ferrara, *Senior Member, IEEE*, and Lalo Magni

Abstract—This paper deals with the design of a robust hierarchical multiloop control scheme to solve motion control problems for robot manipulators. The key elements of the proposed control approach are the inverse dynamics-based feedback linearized robotic multi-input-multi-output (MIMO) system and the combination of a model predictive control (MPC) module with an integral sliding mode (ISM) controller. The ISM internal control loop has the role to compensate the matched uncertainties due to unmodeled dynamics, which are not rejected by the inverse dynamics approach. The external loop is closed relying on the MPC, which guarantees an optimal evolution of the controlled system while fulfilling state and input constraints. The motivation for using ISM, apart from its property of providing robustness to the scheme with respect to a wide class of uncertainties, is also given by its capability of enforcing sliding modes of the controlled system since the initial time instant, allowing one to solve the MPC optimization problem relying on a set of linearized decoupled single-input-single-output (SISO) systems that are not affected by uncertain terms. The proposal has been verified and validated in simulation, relying on a model of a COMAU Smart3-S2 industrial robot manipulator, identified on the basis of real data.

Index Terms—Integral sliding mode (ISM), model predictive control (MPC), robot manipulators, uncertain systems.

I. INTRODUCTION

IN ROBOTICS technology, recent research trends focus on performing particularly critical tasks in an optimal way, while fulfilling some plant constraints in order to avoid failures, wear of the electromechanical parts or to guarantee safe and close robot–human interactions [1], [2]. Yet, typically, industrial robots are controlled by classical potential difference or proportional-integral-derivative (PID) controllers [2], which can fail in guaranteeing this kind of features.

In the last decades, among the control algorithms published in the literature, model predictive control (MPC) represents an appropriate and effective solution to solve this kind of problem, providing an optimal control strategy in case of even complex

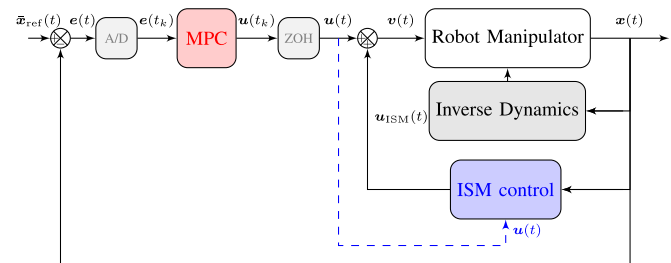


Fig. 1. Scheme of the overall hierarchical multiloop control scheme for robot manipulators.

constrained dynamical systems [3]–[6]. Hence, the online optimization can typically lead to an increased computation time with respect to “classical” control laws. For this reason, MPC has been efficiently used in several industrial processes, such as chemical plants or oil refineries [7], but its application to robotic systems in a true industrial environment, in which unavoidable modeling uncertainties and external disturbances affect the system, is still limited [8]–[11].

Moreover, the MPC method requires the knowledge of the dynamical model of the system, according to which the optimal control sequence is generated by predicting the future evolution of the state variables. Since possible modeling uncertainties can occur, recent research has been devoted to develop robust MPC approaches able to satisfy the system constraints even in these critical cases [12], [13]. In this direction, the two main approaches proposed in the literature are the so-called min–max approach, able to fulfill the plant constraints considering the worst possible uncertainty realization, but at the price of a very high computational burden [14]–[16], and the so-called open-loop nominal approach, where the real constraints are shrunk to guarantee that the original constraints are fulfilled for any possible uncertainty realization [17]–[19].

In this paper, inspired by [20], taking into account the class of robot manipulators, and having the aim of keeping the computational complexity to a minimum, in order to make the proposal really usable in practice, an alternative robust hierarchical multiloop control scheme is proposed (see Fig. 1). More specifically, the control scheme consists of three loops: an inner loop based on the so-called inverse dynamics approach [2], aimed at transforming the nonlinear multi-input-multi-output (MIMO) robotic system into a set of perturbed linearized decoupled single-input-single-output (SISO) systems (the number of systems is equal to the number of the joints of the robot manipulator); a second loop including a controller designed according to the so-called

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G. P. Incremona and A. Ferrara are with the Dipartimento di Ingegneria Industriale e dell’Informazione, University of Pavia, Pavia 27100, Italy (e-mail: gp.incremona@gmail.com; antonella.ferrara@unipv.it).

L. Magni is with the Dipartimento di Ingegneria Civile e Architettura, University of Pavia, Pavia 27100, Italy (e-mail: lalo.magni@unipv.it).

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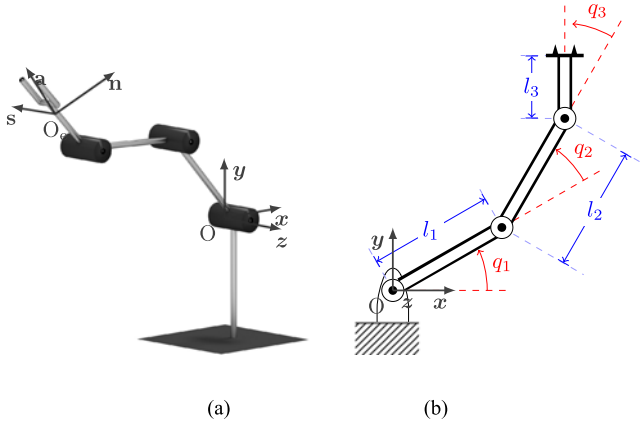


Fig. 2. Anthropomorphic robot manipulator with three joints. (a) spatial schematic view with base-frame and end-effector frame. (b) planar schematic view with joints and links numeration.

integral sliding mode (ISM) control approach [21], which has the role of rejecting at a higher rate all the matched uncertainties [22], [23]; finally, an external loop involving a controller of MPC type with the role of guaranteeing the optimal evolution of the controlled system in the respect of state and input constraints. By the virtue of the linearizing and decoupling effects of the inverse dynamics approach, and of the capability of the ISM controller to make the controlled system insensitive to matched uncertainties since the initial time instant, standard linear MPC methodology [5], running at a slower rate, can be designed in the outer loop, with a clear benefit in terms of containment of computational complexity. A preliminary version of this paper without proofs of the generalized approach and evaluation of the computational costs has been presented in [24].

The present paper is organized as follows. In Section II, the model of the robotic system is introduced, and dynamical aspects are recalled. In Section III, the inverse dynamics approach is described and the control problem to solve is formulated. In Section IV, the proposed control scheme is discussed, illustrating the ISM control component and the MPC control law, whereas in Section V the stability analysis is reported. Section VI is devoted to present simulation results obtained by relying on the model of an industrial manipulator, i.e., a CO-MAU Smart3-S2 anthropomorphic robot. Both the model and the noise used in simulation have been identified on the basis of experimental tests, so that the simulation environment is quite realistic. Finally, keeping in mind an implementation in practical cases, the computational cost of the proposed algorithm is also discussed.

II. ROBOT MODEL

In order to formulate the model of a generic n -joints robot system, dynamical aspects have to be recalled.

For the sake of simplicity, without loss of generality, refer to the three joints ($n = 3$) robot manipulator schematically shown in Fig. 2(b). Let $l_i, i = 1, 2, \dots, n$ denote the length of the i th link, q_1 denote the orientation of the first link with respect to x -axis clockwise positive, and $q_j, j = 2, 3, \dots, n$, denote the displacement of the j th link with respect to the $(j - 1)$ th one

clockwise positive. Let $O = \{x, y, z\}$, denote the base-frame of the robotic manipulator, and $O_e = \{n, s, a\}$ denote the end-effector frame as indicated in Fig. 2(a).

The dynamics of the robot can be written in the joint space, by using the Lagrangian approach, as

$$M(q)\ddot{q} + n(q, \dot{q}) = \tau \quad (1)$$

$$n(q, \dot{q}) = C(q, \dot{q})\dot{q} + F_v\dot{q} + F_s \operatorname{sgn}(\dot{q}) + g(q) \quad (2)$$

where $M(q) \in \mathbb{R}^{n \times n}$ is the inertia matrix, $C(q, \dot{q}) \in \mathbb{R}^{n \times n}$ represents centripetal and Coriolis torques, $F_v \in \mathbb{R}^{n \times n}$ is the viscous friction matrix, $F_s \in \mathbb{R}^{n \times n}$ is the static friction matrix, $g(q) \in \mathbb{R}^n$ is the vector of gravitational torques, and $\tau \in \mathbb{R}^n$ represents the motors torques.

III. PROBLEM FORMULATION

In this section, the considered globally feedback linearized MIMO system will be introduced and the corresponding state-space model will be defined.

A. Inverse Dynamics Control

In order to reduce the nonlinear MIMO robotic system to a linear system, the so-called inverse dynamics approach [2] is used. The inverse dynamics of the robot manipulator can be written, in the joint space, as a nonlinear relationship between the plant inputs and the plant outputs, relying on (1) and (2), so that the control law can be expressed as

$$\tau = M(q)v + \hat{n}(q, \dot{q}) \quad (3)$$

where v is an auxiliary control variable. Typically, the identified $M(q)$ coincides with the actual one, whereas $\hat{n}$ is an estimate of n , which does not necessarily coincide with n [25]. By applying the feedback linearization to system (1) and (2), the resulting model is

$$\ddot{q} = v - \eta(q, \dot{q}) \quad (4)$$

where $\eta(q, \dot{q})$ takes into account the modeling uncertainties and external disturbances, i.e.,

$$\eta(q, \dot{q}) = -M^{-1}(q)(\hat{n}(q, \dot{q}) - n(q, \dot{q})). \quad (5)$$

B. State-Space Model

After the application of the inverse dynamics control, the original MIMO system is reduced to n SISO decoupled systems, one for each joint, in which the state vector is $x_i = [x_{1i} \ x_{2i}]^T = [q_i \ \dot{q}_i]^T$, whereas η_i represents the so-called matched uncertainty [22] such that

$$\begin{cases} \dot{x}_{1i}(t) = x_{2i}(t) \\ \dot{x}_{2i}(t) = v_i(t) - \eta_i(t) \end{cases} \quad (6)$$

which is a double integrator, with $\dot{x}_{2i} = \ddot{q}_i$ being the acceleration of the i th joint.

System (6) can be written in a matrix compact form as the following constrained linear SISO system

$$\dot{x}_i(t) = A_i x_i(t) + B_i(v_i(t) - \eta_i(t)) \quad (7)$$

where $\mathbf{x}_i \in \mathbb{R}^2$ is the state vector, $v_i \in \mathcal{U}$ is the current control variable, with $\mathcal{U} \subset \mathbb{R}$ being a compact set containing the origin point, and $\eta_i \in \mathbb{R}$ the disturbance term of the system. Moreover, $\mathbf{A}_i \in \mathbb{R}^{2 \times 2}$ and $\mathbf{B}_i \in \mathbb{R}^{2 \times 1}$ is full rank. Assume also that the state variables are restricted to fulfill the following constraint

$$\mathbf{x}_i \in \mathcal{X} \quad (8)$$

where $\mathcal{X}$ is a compact set containing the origin as an interior point, whereas the control variable is such that

$$\|v_i\| \leq v_{i_{\max}} \quad (9)$$

with $v_{i_{\max}} = \mathcal{U}^{\sup} := \sup_{v \in \mathcal{U}} \{\|v\|\}$ being the limits of the actuators in terms of acceleration. The uncertainty term η_i is also bounded such that

$$\eta_i \in \mathcal{D} \quad (10)$$

where $\mathcal{D}$ is a compact set containing the origin with known $\mathcal{D}^{\sup} := \sup_{\eta \in \mathcal{D}} \{\|\eta\|\}$.

C. Problem Statement

We are now in a position to be able to formulate the control problem to solve. Given the robot system described in Section II, the control objective is to design a control system in order to make the robot manipulator track a prespecified reference trajectory. In the following section, a functional multirate architecture that combines an MPC module and an ISM controller is designed to solve the aforementioned control problem. The role of the MPC is to make the robot manipulator follow the reference trajectories in an optimal way with respect to input and output constraints. The ISM control rejects matched disturbances and uncertain terms due to unmodeled dynamics, which are not compensated by the inverse dynamics approach.

IV. MPC/ISM CONTROL: THE CONTROL STRATEGY

Making reference to the general control scheme in Fig. 1, the control law is designed on the basis of the feedback linearized system (7). More specifically, the whole auxiliary control variable $v(t)$ is chosen as follows:

$$v(t) = u(t) + u_{\text{ISM}}(t) \quad (11)$$

where u and u_{ISM} are generated by the MPC controller and the ISM controller, respectively. The ISM controller is based on the continuous-time model of the system and on the signal generated by the MPC controller, which, in turn, is based on the discrete-time model of the original system. For the sake of simplicity, in the following sections, we consider only the single SISO system and the subscript i is omitted, when it is obvious.

A. Considered Control Scheme

In Fig. 3, the proposed control scheme is illustrated in detail.

This scheme consists of three control loops. The first loop is based on the inverse dynamics approach, described in Section III. After the inverse dynamics feedback linearization,

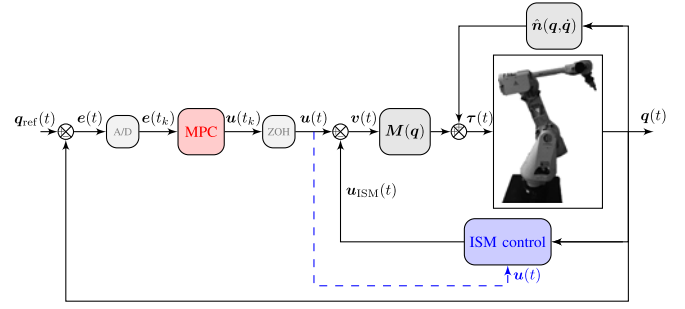


Fig. 3. Detailed MPC/ISM control scheme applied to the considered feedback linearized robot manipulator COMAU Smart3-S2.

the second loop is closed relying on the ISM controller, which computes $u_{\text{ISM}} \in \mathbb{R}^n$ and rejects the matched uncertainty affecting the system. The third loop is designed to implement the MPC based controller, which computes the control $u \in \mathbb{R}^n$ combined with u_{ISM} so as to solve the reference tracking problem in an optimal way while satisfying the constraints. The position error of the controlled system, given as input to the MPC module, is defined as $e = q_{\text{ref}} - q$, q_{ref} being the desired reference trajectory.

B. ISM Controller

The ISM control has the feature to provide robustness to the scheme in front of a wide class of uncertainties, and to enforce sliding modes of the controlled system since the initial time instant. This control approach requires the knowledge of a nominal model of a system that can be also nonlinear, a properly designed high level control law (MPC in our case), and a discontinuous control action in order to remove the uncertain terms. Considering the dynamic system (6), assume that, for each joint, the so-called integral sliding variable $\sigma_i \in \mathbb{R}$ (see [21]) is defined as follows:

$$\sigma_i(\mathbf{x}_i(t)) = \mathbf{S}_i \left(\mathbf{x}_i(t) - \mathbf{x}_i(t_0) - \int_{t_0}^t [\mathbf{x}_{2i}(\zeta), u_i(\zeta)]^T d\zeta \right) \quad (12)$$

with $\sigma_i(\mathbf{x}_i(t)) = 0$ being the associated integral sliding manifold, t_0 the initial time instant, $\mathbf{S}_i$ the row vector $[c_i, 1]$, and c_i a positive constant.

Now, the control law can be expressed as follows:

$$u_{\text{ISM}} = -U_{i_{\max}} \text{sgn}(\sigma_i) \quad (13)$$

where $U_{i_{\max}} > \mathcal{D}^{\sup}$ is suitably chosen in order to enforce the sliding mode, with $\mathcal{D}^{\sup}$ depending on the modeling uncertainties and disturbances (5).

Remark 1: Note that ISM control can imply the so-called chattering phenomenon [26], which can be avoided by using integral higher order sliding modes as that in [27]. ■

Having in mind a robotic application, as shown in [21], an effective solution, able to cope with chattering and avoid high-frequency switching of the variable structure component of the control torque, consists in using the so-called equivalent control [22]. The equivalent control cannot be computed, since it depends on the uncertain terms affecting the system. However,

in [21], it is shown that an approximation of the equivalent control can be obtained via a first-order linear filter with the real discontinuous control (13) as input signal, i.e.,

$$\tilde{u}_{i\text{ISM}_{\text{eq}}}(t) = \frac{1}{\mu_i} \int_{t_0}^t e^{-\frac{1}{\mu_i}(t-\zeta)} u_{i\text{ISM}}(\zeta) d\zeta \quad (14)$$

where μ_i is the time constant of the filter, whereas the sliding variable has to be redesigned as

$$\begin{aligned} \sigma_i(\mathbf{x}_i(t)) = \\ \mathbf{S}_i \left(\mathbf{x}_i(t) - \mathbf{x}_i(t_0) - \int_{t_0}^t [x_{2_i}(\zeta), v_i(\zeta) - u_{i\text{ISM}}(\zeta)]^T d\zeta \right) \end{aligned} \quad (15)$$

with $v_i(t) = u_i(t) + \tilde{u}_{i\text{ISM}_{\text{eq}}}(t)$. Note that, since a first-order linear filter is used to obtain the equivalent control, the sliding variable has to be modified as in (15) so to take into account the difference between the filtered control (14) and the discontinuous one (13), intrinsically compensating the mismatch between the actual equivalent control $u_{i\text{ISM}_{\text{eq}}}$ and its average value $\tilde{u}_{i\text{ISM}_{\text{eq}}}$. This implies that the sliding manifold is continuously adapted in order to guarantee the ideal sliding mode generation $\forall t \geq t_0$.

The effect of the ISM control law is that of rejecting the uncertainty of system (6) so as to obtain

$$\begin{cases} \dot{x}_{1_i}(t) = x_{2_i}(t) \\ \dot{x}_{2_i}(t) = u_i(t) \end{cases} \quad (16)$$

that is

$$\dot{\mathbf{x}}_i(t) = \mathbf{A}_i \mathbf{x}_i(t) + \mathbf{B}_i u_i(t) \quad (17)$$

which is a double integrator without disturbances affecting the system. Note that the ISM control cannot violate the state constraints due to the fact that the sliding variable also depends on the MPC control law.

Remark 2: As suggested in [21], the time constant μ_i should be set such that the fundamental component of the discontinuous action is not distorted. ■

C. Model Predictive Controller

By virtue of the rejection of matched uncertainties and external disturbances through the use of the ISM controller, the MPC controller can be designed starting from the nominal system, i.e., the dynamical system without matched uncertainties. For this reason, it is not necessary to consider conservative robust MPC approaches [16], [19], but only a nominal MPC is required, with a significant beneficial effect in terms of computational burden. In the following, for the sake of simplicity, the subscript i will be omitted, when obvious.

The MPC controller is designed on the discrete time system of (17), i.e.,

$$\mathbf{x}_i(t_{k+1}) = \tilde{\mathbf{A}}_i \mathbf{x}_i(t_k) + \tilde{\mathbf{B}}_i u_i(t_k). \quad (18)$$

Note that, for the design of a continuous time MPC and a rigorous proof of stability, one can refer to [28]. The adopted

MPC controller is based on the solution of the so-called finite-horizon optimal control problem (FHOC). The latter consists in minimizing, at any sampling time t_k , a suitably defined cost function with respect to the control sequence $\mathbf{u}_{[t_k, t_k+N-1|t_k]} := [u_0(t_k), u_1(t_k), \dots, u_{N-1}(t_k)]$, where $N \geq 1$ is the prediction horizon. In our case, the cost function to minimize with respect to $\mathbf{u}_{[t_k, t_k+N-1|t_k]}$ is a quadratic function as

$$\begin{aligned} J(\mathbf{e}_i(t_k), \mathbf{u}_{[t_k, t_k+N-1|t_k]}, N) = \\ \sum_{j=0}^{N-1} \|\mathbf{e}_i(t_{k+j})\|_{\mathbf{Q}_i}^2 + \|u_i(t_{k+j})\|_{R_i}^2 + \|\mathbf{e}_i(t_{k+N})\|_{\mathbf{\Pi}_i}^2 \end{aligned} \quad (19)$$

where the notation $\|\cdot\|_{\mathbf{W}}^2$ stands for the square norm of a vector weighted by a matrix $\mathbf{W}$.

The cost function (19) is subject to the hard constraints represented by the dynamics of system (18), and inequalities constraints on states and input variables, i.e.,

$$\mathbf{x}_i(t_{k+j}) \in \mathcal{X} \quad (20)$$

$$\mathbf{x}_i(t_{k+N}) \in \mathcal{X}_f \quad (21)$$

$$\|u_i(t_{k+j})\| \leq v_{i\max} - U_{i\max} \quad (22)$$

with $j = 1, \dots, N-1$. Moreover, $\mathcal{X}_f$ is the so-called terminal set such that $\mathbf{x}_i(t_{k+N}) \in \mathcal{X}_f$, with

$$\mathcal{X}_f := \{\mathbf{x}_i \mid \|\mathbf{x}_i - \bar{\mathbf{x}}_{i\text{ref}}\|_{\mathbf{\Pi}}^2 \leq \rho\}, \quad \mathcal{X}_f \subseteq \mathcal{X} \quad (23)$$

for any constant $\bar{\mathbf{x}}_{i\text{ref}} \in \mathcal{X}_{\text{ref}}$ such that $(\bar{\mathbf{x}}_{i\text{ref}}, 0)$ is an equilibrium point for systems (18) and with $\mathcal{X}_f$ containing the origin as an interior point. In order to define the terminal set and the terminal penalty, one needs to introduce an auxiliary control law $\kappa_{i\text{f}}(\mathbf{e}_i)$ that can be

$$\kappa_{i\text{f}}(\mathbf{e}_i(t_k)) = \mathbf{K}_{\text{LQ}} \mathbf{e}_i(t_k) \quad (24)$$

$\mathbf{K}_{\text{LQ}}$ being the control gain of an infinite horizon Linear Quadratic (LQ) controller with the same cost function. Note that, the value ρ in (24) is a positive real number such that $\forall \mathbf{x}_i(t_{\bar{k}}) \in \mathcal{X}_f, \forall t_k > t_{\bar{k}}$ it yields

$$\mathbf{x}_i(t_k) \in \mathcal{X}_f \quad (25)$$

$$\|\kappa_{i\text{f}}(\mathbf{e}_i(t_k))\| \leq v_{i\max} - U_{i\max} \quad (26)$$

with $\mathbf{e}_i(t_k) = \mathbf{x}_i(t_k) - \bar{\mathbf{x}}_{i\text{ref}}(t_k)$. In (19), $\mathbf{Q}_i$ is a positive definite matrix, R_i is a scalar weight, and $\mathbf{\Pi}_i$ is the positive definite terminal state weight associated with the terminal penalty $V_f = \|\mathbf{e}_i(t_{k+N})\|_{\mathbf{\Pi}_i}^2$, which is assumed such that

$$V_f(\mathbf{e}_i(t_{k+1})) - V_f(\mathbf{e}_i(t_k)) + \|\mathbf{e}_i(t_k)\|_{\mathbf{Q}_i}^2 + \|\kappa_{i\text{f}}(\mathbf{e}_i(t_k))\|_{R_i}^2 \leq 0 \quad (27)$$

so as to ensure the stability of the controlled system. The matrix $\mathbf{\Pi}_i$ represents instead the solution of the Riccati equation

$$(\tilde{\mathbf{A}}_i - \tilde{\mathbf{B}}_i \mathbf{K}_{\text{LQ}})^T \mathbf{\Pi}_i (\tilde{\mathbf{A}}_i - \tilde{\mathbf{B}}_i \mathbf{K}_{\text{LQ}}) - \mathbf{\Pi}_i = -\mathbf{Q}_i - \mathbf{K}_{\text{LQ}}^T R_i \mathbf{K}_{\text{LQ}}. \quad (28)$$

Then, according to the receding-horizon strategy, the applied piecewise-constant control law is the following:

$$u_i(t) = \kappa_{\text{MPC}}(\mathbf{e}_i(t_k)), \quad t \in [t_k, t_{k+1}) \quad (29)$$

where $t_{k+1} - t_k = T$ is the MPC sampling time, and

$$\kappa_{\text{MPC}}(e_i(t_k)) := u_{i0}^o(t_k) \quad (30)$$

with $u_{i0}^o(t_k)$ is the first value at t_k of the optimal control sequence for the i th joint, obtained by solving the FHOC.

V. STABILITY ANALYSIS

With reference to the proposed control approach, the following results can be proved.

Lemma 1: Given the MIMO nonlinear robotic model (1) and (2), by applying the inverse dynamics approach in (3), one obtains n decoupled perturbed double integrators. ■

Proof. The proof of this lemma in case of an ideal compensation is reported in [2]. Since unavoidable modeling uncertainties are present, applying the inverse dynamics (3) to (1), one obtains

$$\ddot{q} = v + M^{-1}(q)(\hat{n}(q, \dot{q}) - n(q, \dot{q})) = v - \eta(q, \dot{q}) \quad (31)$$

with the uncertain terms such that (5) holds. Hence, the result is a set of n decoupled perturbed double integrators as

$$\begin{cases} \dot{x}_{1i}(t) = x_{2i}(t) \\ \dot{x}_{2i}(t) = v_i(t) - \eta_i(t) \end{cases} \quad (32)$$

with $x_{1i} = q_i$ and $x_{2i} = \dot{q}_i$, $i = 1, \dots, n$, which concludes the proof. ■

Lemma 2: Given system (7), controlled via (11), (14), and (30), with the sliding variable (15) such that the matrix $S_i B_i$ is nonsingular, then an ISM is enforced on the integral sliding manifold $\sigma_i = 0$, $\forall t \geq t_0$. ■

Proof. The proof directly follows from [21] (see also [29, Ch. 7] for further details), according to which the fact that an ISM is enforced since the initial time instant t_0 can be straight forward using the Lyapunov second method by considering the function $V_{\text{ISM}_i} = 0.5\sigma_i^2$ as the Lyapunov candidate. ■

In the following theorem, the equivalent system controlled by the MPC component will be obtained and the role of the ISM component as *perturbation estimator* will be exploited.

Theorem 1: Given system (7), controlled via (11), (14), and (30), then $\forall t \geq t_0$ the equivalent system results in being

$$\dot{x}_i(t) = A_i x_i(t) + B_i u_i(t)$$

with $i = 1, \dots, n$. ■

Proof. By virtue of Lemmas 1 and 2, since the initial time instant t_0 one has that $\sigma_i = 0$ so that also its time derivative is $\dot{\sigma}_i = 0$, i.e.,

$$\begin{aligned} \dot{\sigma}_i &= S_i \dot{x}_i - S_i [x_{2i} \ u_i + \tilde{u}_{i\text{ISM}_{\text{eq}}} - u_{i\text{ISM}}]^T \\ &= c_i x_{2i} + v_i - \eta_i - c_i x_{2i} - u_i - \tilde{u}_{i\text{ISM}_{\text{eq}}} + u_{i\text{ISM}} \\ &= u_i + \tilde{u}_{i\text{ISM}_{\text{eq}}} - \eta_i - u_i - \tilde{u}_{i\text{ISM}_{\text{eq}}} + u_{i\text{ISM}} = 0. \end{aligned} \quad (33)$$

According to the *equivalent control* concept (see [22] for a definition), one has that in Filippov sense

$$0 = u_i + \tilde{u}_{i\text{ISM}_{\text{eq}}} - \eta_i - u_i - \tilde{u}_{i\text{ISM}_{\text{eq}}} + u_{i\text{ISM}_{\text{eq}}} = u_{i\text{ISM}_{\text{eq}}} - \eta_i.$$

Hence, one can conclude that $u_{i\text{ISM}_{\text{eq}}} = \eta_i$. Substituting the latter expression into system (7), controlled via (11), the equivalent system results in being (17), which proves the theorem. ■

Starting from Lemmas 1 and 2 and Theorem 1, the following theorem can be proved.

Theorem 2: Given system (18), controlled via (30), obtained by solving the FHOC with cost function (19) subject to the system dynamics, input and state constraints (21)-(22), then, $x_i = \bar{x}_{i\text{ref}}, \forall i = 1, \dots, n$, results in being an asymptotically stable equilibrium point of the controlled system. ■

Proof. The proof of the theorem has to be carried out in two steps. First, it is necessary to prove the recursive feasibility, i.e., given an optimal solution at time t_k , it is always possible to find a solution at time t_{k+1} that satisfies all the constraints.

Step 1 (Feasibility) Consider the optimal solution given at t_k , $\mathbf{u}_{i[t_k, t_{k+N-1}|t_k]}^o := [u_{i0}^o(t_k), u_{i1}^o(t_k), \dots, u_{iN-1}^o(t_k)]$. According to the receding-horizon principle, only the first element of the optimal sequence is applied. At the time instant t_{k+1} the control sequence

$$\tilde{\mathbf{u}}_{i[t_{k+1}, t_{k+N}|t_{k+1}]} = \begin{cases} \mathbf{u}_{i[t_{k+1}, t_{k+N-1}|t_k]}^o \\ \kappa_{i_t}(e_i(t_{k+N})) \end{cases} \quad (34)$$

fulfills the constraints (21)–(23). In fact, since it holds $\tilde{\mathbf{u}}_{i[t_{k+1}, t_{k+N-1}|t_{k+1}]} = \mathbf{u}_{i[t_{k+1}, t_{k+N-1}|t_k]}^o$, constraints (21) and (23) are fulfilled. Moreover, from (22), it holds that $x_i(t_{k+N}) \in \mathcal{X}_f$. Hence, from (27), it also holds that $\|\kappa_{i_t}(e_i(t_{k+N}))\| \leq v_{i\text{max}} - U_{i\text{max}}$ and $x_i(t_{k+N}) \in \mathcal{X}$, so that (21) and (23) are satisfied also for $j = N$. Finally, from (26), it follows that $x_i(t_{k+N+1}) \in \mathcal{X}_f$, which concludes the proof of the feasibility.

After having proved the recursive feasibility, the second step is to prove the stability properties of the system controlled via the MPC law.

Step 2 (Stability) In order to prove the asymptotical stability, we need to find a Lyapunov function candidate. We chose the function $J^o(e_i(t_k), t_k) > 0, \forall e_i \neq \mathbf{0}$, and $J^o(\mathbf{0}, t_k) = 0$, associated with the cost function (19). Consider the cost function $\tilde{J}(e_i(t_{k+1}), t_{k+1})$ associated with the feasible control sequence (35). Since, this function is not a priori the optimal one, it holds $J^o(e_i(t_{k+1}), t_{k+1}) \leq \tilde{J}(e_i(t_{k+1}), t_{k+1})$. From (19), by using (28), one has that

$$\begin{aligned} &\tilde{J}(e_i(t_{k+1}), t_{k+1}) - J^o(e_i(t_k), t_k) \\ &= -\|e_i(t_k)\|_{Q_i}^2 - \|\kappa_{i_t}(e_i(t_k))\|_{R_i}^2 + V_f(e_i(t_{k+N+1})) + \\ &\quad - V_f(e_i(t_{k+N})) + \|e_i(t_{k+N})\|_{Q_i}^2 + \|\kappa_{i_t}(e_i(t_{k+N}))\|_{R_i}^2 \\ &< -\|e_i(t_k)\|_{Q_i}^2 - \|\kappa_{i_t}(e_i(t_k))\|_{R_i}^2 < 0 \end{aligned} \quad (35)$$

which implies that J^o is a decreasing function. Then, one can conclude that $x_i = \bar{x}_{i\text{ref}}, \forall i = 1, \dots, n$, results in being an asymptotically stable equilibrium point of the controlled system, which concludes the proof. ■

VI. CASE STUDY

The robotic system we are dealing with is a 6-joint robot manipulator. For the sake of simplicity, we consider only vertical planar motions of the robotic manipulator, locking three of the six joints of the robot. However, the proposed control scheme and the design of the controllers could have a more general validity, even in the spatial case for 6-joint robot manipulators.

TABLE I
STATE AND INPUT CONSTRAINTS FOR EACH JOINT

Joint i	$q_{i\max}$ (rad)	$\dot{q}_{i\max}$ (rad/s)	$v_{i\max}$ (rad/s ²)
1	1.83	2	145
2	2.71	3.5	250
3	3.49	6.3	350

TABLE II
PERFORMANCE INDEXES FOR EACH JOINT

Strategy	Joint i	$e_{\text{RMS}i}$ (rad)	E_{c_i} (rad/s ²)
MPC	1	0.2105	8.0303
	2	0.3749	13.4839
	3	1.0458	41.2172
MPC/ISM	1	0.0070	8.0600
	2	0.0102	13.4217
	3	0.0152	34.8530

The control strategy, previously proposed, has been applied in simulation to the model of a COMAU Smart3-S2 anthropomorphic industrial robot by using the software MATLAB Simulink. Note that the model has been identified on the basis of real data through experimental tests [25], so that the simulation environment is quite realistic.

The simulation scenario has also been made more realistic by injecting the disturbance terms $\eta = [\eta_1, \eta_2, \eta_3]^T$ to the acceleration of the joints on the basis of the effective disturbances registered during experimental tests [27]. These disturbances represent modeling uncertainties, such as friction, centripetal or Coriolis forces, which are not completely compensated by the inverse dynamics control (3). The corresponding bounds of the uncertainties for joints 1, 2 and 3 are 20, 30 and 80 rad/s², respectively. Moreover, according to the mechanical limits of the real robot, the position, velocity, and acceleration constraints are those reported in Table I. Note that these values are the same provided by the robot manufacturer and imposed by the limit switches fastened on the real system. The initial conditions of the joint variables are $q_0 = [0, 0, 0]^T$, whereas the target position is $q_{\text{ref}} = [\pi/4, \pi/3, 2\pi/4]^T$.

In order to show the effectiveness of the control law (11), the latter is compared with the case in which a nominal MPC without ISM control is applied to the robot manipulator. The sliding variable has been chosen as in (15) with $c_i = 10$, $i = 1, 2, 3$, the ISM control is as in (14), while the ISM control gains are 20, 35, and 85, respectively. The MPC parameters have been chosen such that for each joint $Q_i = \text{diag}(100, 100)$, $R_i = 0.1$, and the terminal weight equal to

$$\Pi_i = \begin{bmatrix} 5213.4 & 165.8 \\ 165.8 & 221.3 \end{bmatrix}. \quad (36)$$

Moreover, the sampling time of the simulation has been set as in the real case equal to $t_s = 0.001$ s, whereas the sampling time of the MPC loop has been set as $T = 0.02$ s, with prediction horizon $N = 10$.

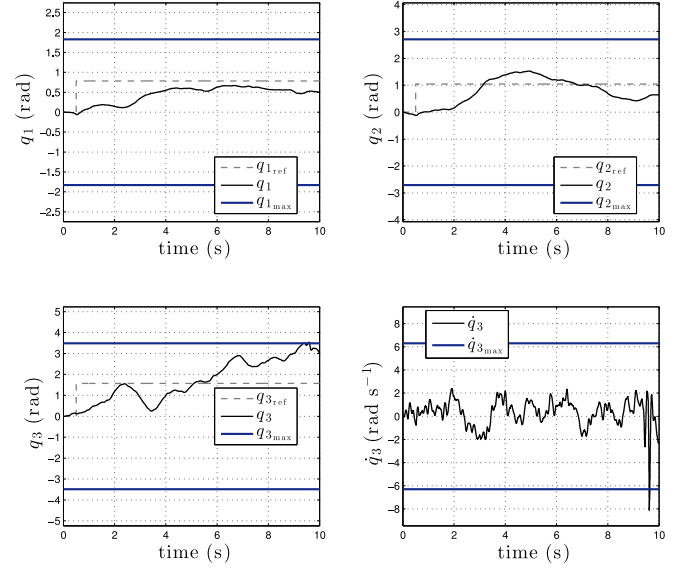


Fig. 4. Time evolution of the joint variables for each joint, and the velocity $\dot{q}_3$ when only the nominal MPC is used.

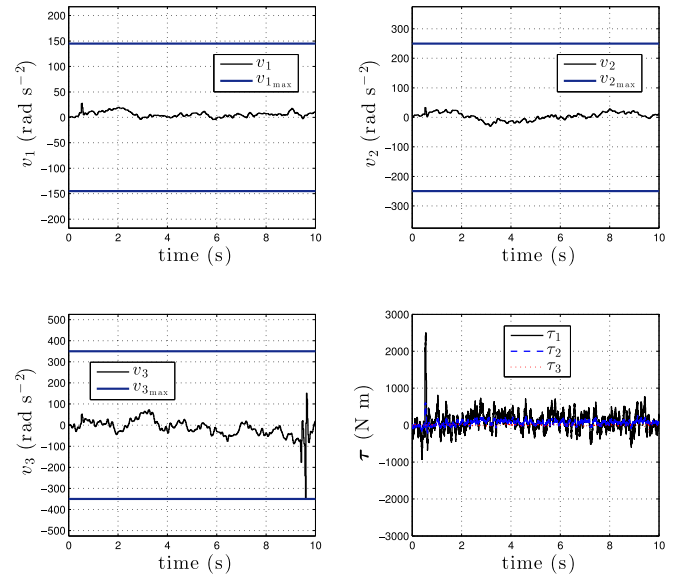


Fig. 5. Time evolution of the auxiliary control variables and of the torque for each joint when only the nominal MPC is used.

A. Results and Comparison

The tracking control performance is evaluated through the following indexes: 1) the root mean square (RMS) error (e_{RMS}); and 2) the RMS value of the control action (E_c).

Table II shows the outcome indexes achieved through the MPC standalone and the proposed MPC/ISM strategy.

The RMS error in steady-state is significantly smaller when the MPC/ISM is used for all the joints with respect to the case with only MPC. This is clearly represented in Fig. 4 where the uncertainty strongly affects the position tracking performance and the state constraints are violated (see position and velocity of joint 3), when only MPC is used. Fig. 5 shows the

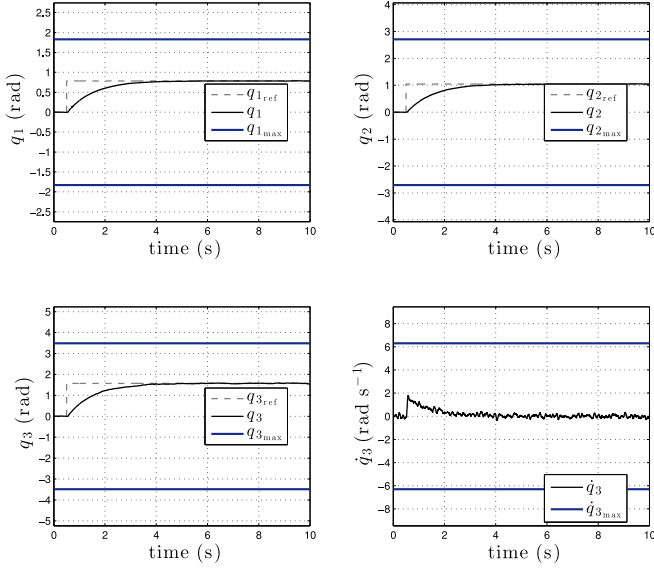


Fig. 6. Time evolution of the joint variables for each joint, and the velocity $\dot{q}_3$ when the MPC/ISM strategy is used.

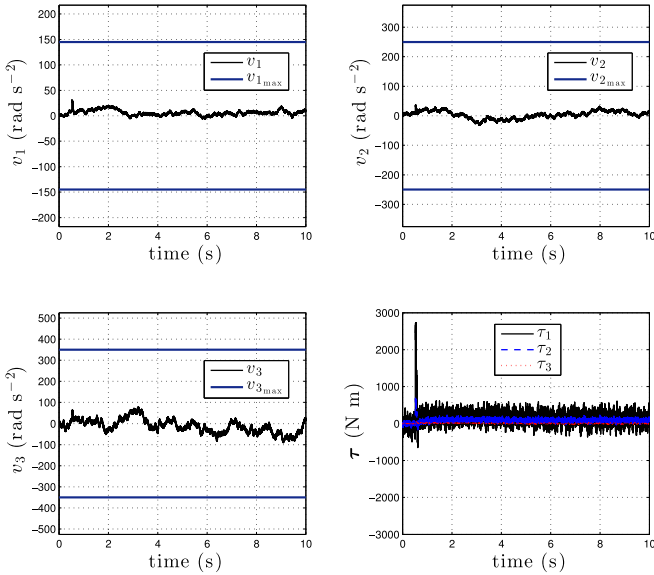


Fig. 7. Time evolution of the auxiliary control variables and of the torque for each joint when the MPC/ISM strategy is used.

corresponding auxiliary control variables and the torques directly fed into the plant. On the other hand, a very precise tracking with constraints satisfaction can be observed in Fig. 6 when the MPC/ISM control is applied. The corresponding auxiliary control variables and torques are illustrated in Fig. 7. This beneficial effect is given by the ISM component that perfectly estimates and rejects the uncertain terms affecting the systems (see Fig. 8). As for the control effort, it results in being very similar in both cases, except for Joint 3 for which it is smaller in the case of MPC/ISM control, as can be observed in the graphical rendering of the performance indexes, normalized with respect to the worse cases, as shown in Fig. 9.

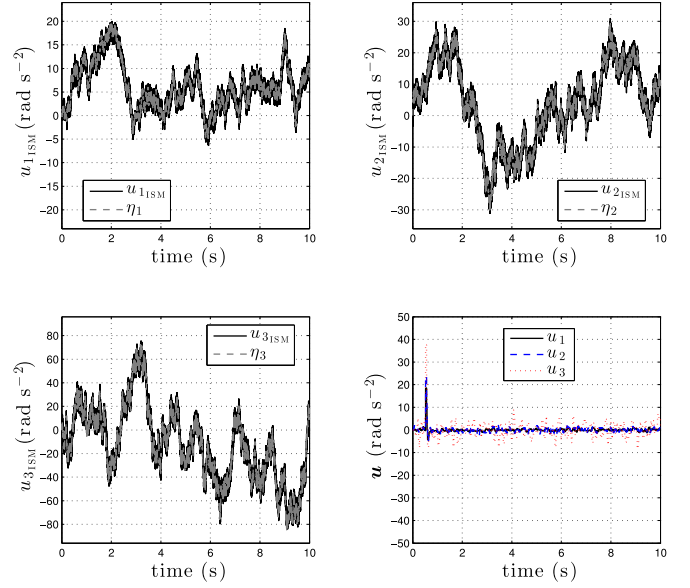


Fig. 8. Time evolution of the ISM control component with respect to the matched uncertainty affecting the system, and the MPC component when the MPC/ISM strategy is used.

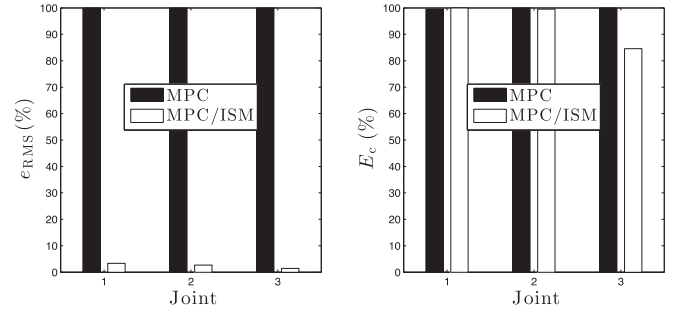


Fig. 9. Performance indexes normalized with respect to the worse case.

TABLE III
TIME CONSUMPTION OF THE PROPOSED CONTROL STRATEGY IN SECONDS

Algo.	Mean	Min.	Max.	Std. dev.
MPC	0.018	0.017	0.71	0.011
ISM	2.9×10^{-5}	2.7×10^{-5}	0.0086	1.4×10^{-4}

B. Evaluation of Application

To evaluate the computational costs of the proposed control strategy in practical robotic cases, an implementation in MATLAB code, with quadprog as MPC solver, running on a laptop-computer with an Intel Core 2 Duo at 2.4GHz with 4-GB RAM, has been performed. The time measurement has been made over four thousand executions of each algorithm. This can be considered as a conservative upperbound of a practical implementation made with any imperative assembly language. Table III reports the results expressed in terms of mean, minimum, maximum,

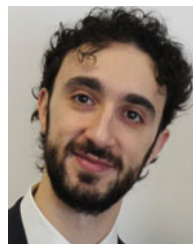
and standard deviation of the execution time. Their respective sum for each algorithm is an estimation of the total execution time. As expected the higher computational burden is required by the MPC component, but by virtue of the reduction of the complexity of the optimization problem relying on a simple constrained linear system, it results in being reasonable to be implemented in any standard recent robot control unit or embedded systems such as field programmable gate array (FPGA). As for the ISM component, the obtained execution time confirms its computational lightweight, which makes ISM control a powerful easy-to-implement solution even in an industrial field.

VII. CONCLUSION

In this paper a hierarchical multiloop control scheme based on the combination of MPC and ISM control has been proposed to solve motion control problems for robot manipulators. A basic inverse dynamics feedback linearizing approach is applied to obtain a set of linearized decoupled SISO systems. The ISM control, which runs at higher rate, makes the systems insensitive to the matched uncertainties presence. Finally, the external loop is characterized by the MPC component with the aim to ensure the optimal evolution of the controlled system in the respect of state and input constraints, while keeping the computational complexity to a minimum. This makes the proposal really usable in practice, as also verified by evaluating the possible computational costs of the involved algorithms. The proposed control scheme has been validated in simulation relying on a realistic model of an industrial COMAU Smart3-S2 robot manipulator, identified on the basis of experimental tests.

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Gian Paolo Incremona (M'10) received the bachelor's and master's degrees (Cum Laude) in electric engineering, and the Ph.D. degree in electronics, electric, and computer engineering from the University of Pavia, Pavia, Italy, in 2010, 2012, and 2016, respectively. He was a student of the Almo Collegio Borromeo of Pavia, Pavia, and of the class of Science and Technology of the Institute for Advanced Studies, Istituto Universitario di Studi Superiori di Pavia, Pavia.

He is currently a Postdoc with the Identification and Control of Dynamic Systems Laboratory, University of Pavia. From October 2014 to December 2014, he was in the Dynamics and Control Group, Eindhoven Technology University, Eindhoven, The Netherlands. His research interests include nonlinear control, variable structure control of sliding mode type, model predictive control, networked control, industrial robotics, power systems, and glycemia control in diabetic subjects.



Antonella Ferrara (SM'03) received the laurea degree (Cum Laude) in electronic engineering, and the Ph.D. degree in computer science and electronics from the University of Genova, Genova, Italy, in 1987 and 1992, respectively.

She is currently a Full Professor of automatic control with the University of Pavia, Pavia, Italy. She has authored/co-authored more than 300 papers, including more than 90 journal papers. Her research interests include sliding mode and nonlinear control with application to traffic, auto-

motive, and robotics.

Prof. Ferrara was the Associate Editor of the IEEE TRANSACTIONS ON CONTROL SYSTEMS TECHNOLOGY and the IEEE TRANSACTIONS ON AUTOMATIC CONTROL. Since January 2014, she has been the Associate Editor of IEEE CONTROL SYSTEMS MAGAZINE. She is a Senior Member of the IEEE Control Systems Society, and among others, a member of the Technical Committee on Variable Structure and Sliding Mode Control, and the Technical Committee on Transportation Systems. Since July 2013, she is the Chair of the Women in Control Standing Committee of the Control Systems Society.



Lalo Magni received the laurea degree (Cum Laude) in computer engineering, and the Ph.D. degree in computer science and electronics from the University of Pavia, Pavia, Italy, in 1994 and 1998, respectively.

He is currently a Full Professor of automatic control with the University of Pavia, Pavia, Italy. From October 1996 to February 1997 and in March 1998, he was at the Center for Systems Engineering and Applied Mechanics (CESAME), Université Catholique de Louvain,

Louvain La Neuve, Belgium. From October to November 1997, he was at the University of Twente, Enschede, The Netherlands, with the System and Control Group in the Faculty of Applied Mathematics. He was Plenary, Semiplenary, or Keynote Speaker at several international conferences. His current research interests include nonlinear control, predictive control, receding-horizon control, robust control, process control, and artificial pancreas. His research is witnessed by more than 70 papers published in the main international journals.

Prof. Magni was the Associate Editor of the IEEE TRANSACTIONS ON AUTOMATIC CONTROL and *Automatica*. In September 2008, he was Chair of the Nonlinear Model Predictive Control (NMPC) Workshop on Assessment and Future Direction, Pavia. He was the National Principal Investigator of a Fondo per gli Investimenti della Ricerca di Base (FIRB) Project Futuro in Ricerca and the Local Investigator of an FP7 UE Program.